

# FREEDOM INTERNATIONAL SCHOOL

## WORKSHEET

### PHYSICS

#### CLASS XI

#### GRAVITATION

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1. Can we determine the gravitational mass of a body inside an artificial satellite?
2. A person sitting in a satellite feels weightlessness but a person standing on moon has weight though moon is also a satellite of the earth. Give reason.
3. A satellite is orbiting the earth with a speed  $v_0$ . To make the satellite escape, what should the minimum percentage increase in its velocity be?
4. The radii of two planets are  $R$  and  $2R$  respectively and their densities  $\rho$  and  $\rho/2$  respectively. What is the ratio of acceleration due to gravity at their surfaces?
5. A simple pendulum has a time period  $T_1$  when on the earth's surface, and  $T_2$  when taken to a height  $R$  above the earth's surface, where  $R$  is the radius of the earth. What is the value of  $T_2/T_1$ ?
6. The escape velocity of a projectile on the earth's surface is  $11.2 \text{ km/s}$ . A body is projected out with thrice this speed. What is the speed of the body far away from the earth? Ignore the presence of the sun and other planets.
7. A satellite orbits the earth at a height of  $400 \text{ km}$  above the surface. How much energy must be expended to rocket the satellite out of the earth's gravitational influence?  
Mass of the satellite =  $200 \text{ kg}$ , mass of the earth =  $6.0 \times 10^{24} \text{ kg}$ ,  
radius of earth =  $6.4 \times 10^6 \text{ m}$ .
8. A geostationary satellite orbits the earth at a height of nearly  $36,000 \text{ km}$  from the surface of the earth. What is the potential due to earth's gravity at the site of the satellite?
9. If earth has a mass 9 times and radius twice that of a planet Mars, calculate the minimum velocity required by a rocket to pull out of gravitational force of Mars. Take the escape velocity on the surface of earth to be  $11.2 \text{ km/s}$ .
10. A particle is projected upward from the surface of the earth (radius  $R$ ) with a K.E. equal to half the minimum value needed for it to escape. To what height does it rise above the surface of the earth?